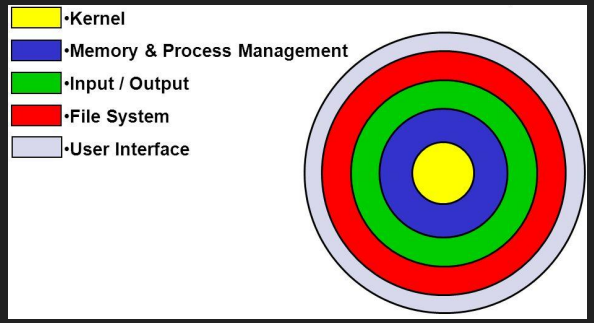
Task 1 - Explain the purpose of operating systems

Leandro Baroca

Functions of Windows Operating Systems

Kernel



In an operating system, the kernel is basically the core of the operating system, and uses interprocess communication and system calls. What I mean by this is that it acts like a bridge between applications and the data processing performed at the hardware level.

If an operating system is being loaded into the memory, the kernel loads first and remains in memory until the operating system is shut down again.

The kernel is responsible for many low-level tasks, such as: disk management, task management, memory management, process management for application execution, device management through use of device drivers and the system call control, which is essential for the executions of kernel services.

User Interface

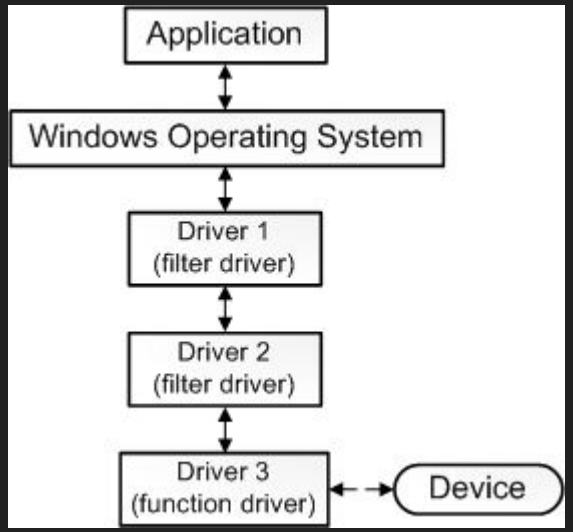


The user interface, or “UI”, is the software which a person can control such as a software application or a hardware device. Software programs have a GUI, or graphical user interface. What I mean by this is that the program has graphical controls, which gives the user the ability to use a mouse or keyboard.

It is important that the user interface gives a user friendly experience to the user because it gives the user a much more natural and smooth way of using the software or hardware. A UI which isn't user friendly or overly complicated will be seen as a badly made UI.

A UI does not only apply to software, hardware devices can also have a user interface, hardware UI is a lot simpler than the more complex software interface. An example of a hardware UI is a remote control for a TV.

Device Drivers

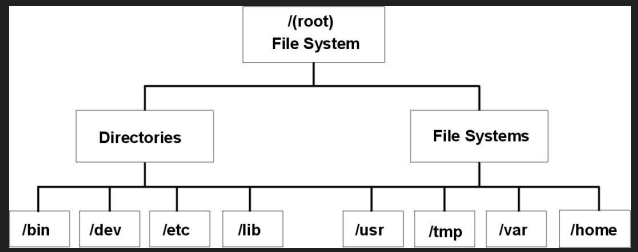


The device driver is basically a small program that enables a computer and a device to communicate with each other. Operating systems on a computer usually comes with preinstalled device drivers. However, the device driver are compatible with the the popular devices on the time it was installed or updated.

Device drivers don't have an automatic update, so that means that the older or newer versions of drivers have to be manually installed by the user. Some Windows programs are virtual device drivers, which is when the programs interface with the Windows Virtual Machine Manager.

In a Windows operating system, a device driver file usually has a file name suffix of DLL or EXE. A virtual device driver usually has the suffix of VXD.

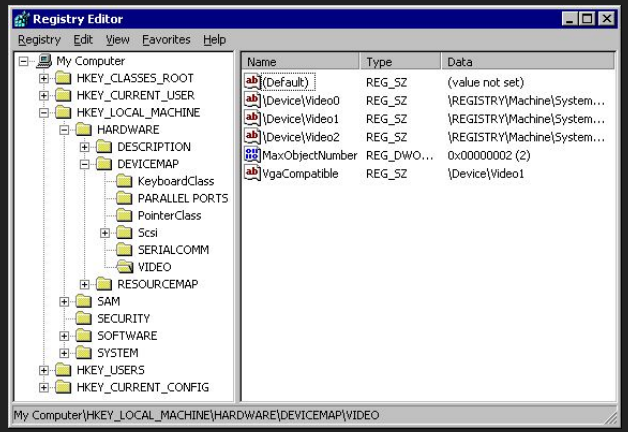
File System Management



The file system management is a system of an operating system or a program that manages the data files within a computer system. The system is designed to manage individual or group files such as special office documents and records, but the system has very limited capabilities. What I mean by this is that the system may contain simple features such as: Assigning queued document numbers for processing, owner and process mapping to track various stages of processing, report generation, notes, status, create, modify, copy, delete and other file operations.

File management should not be confused with a file system, which manages all types of data and files in an OS. Also with database management system (DBMS), which has a relational database capability and includes a programming language for data manipulation

Registry



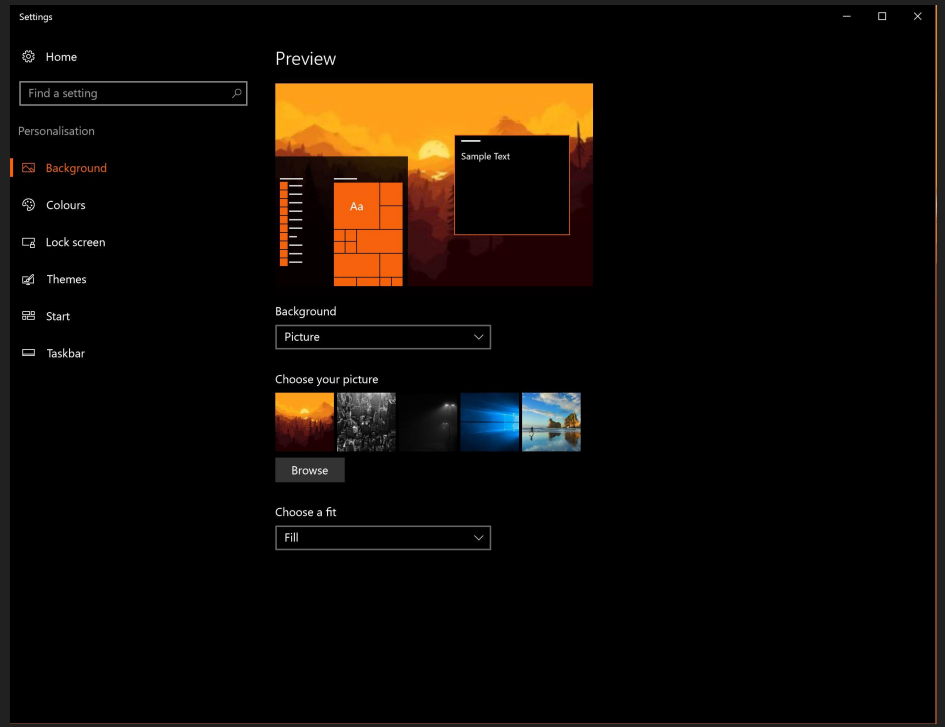
The registry on the Windows OS is used to store much of the information and settings for software programs, hardware devices, user preferences, operating system configurations and much more.

For example, if a new program is installed, new sets of instructions and file references can be added into the registry in a specific location for the program. It is thought that the registry is a bit like the DNA for the Windows operating system.

Of course, the Windows applications rely on Windows Registry, but not all applications necessarily need to utilize the Windows Registry. Some programs store their configurations within XML files instead of the registry. Other applications are entirely portable and store their data in an executable file.

Features of Windows 10 Operating System

Customizing the User Interface (Desktop, etc.)

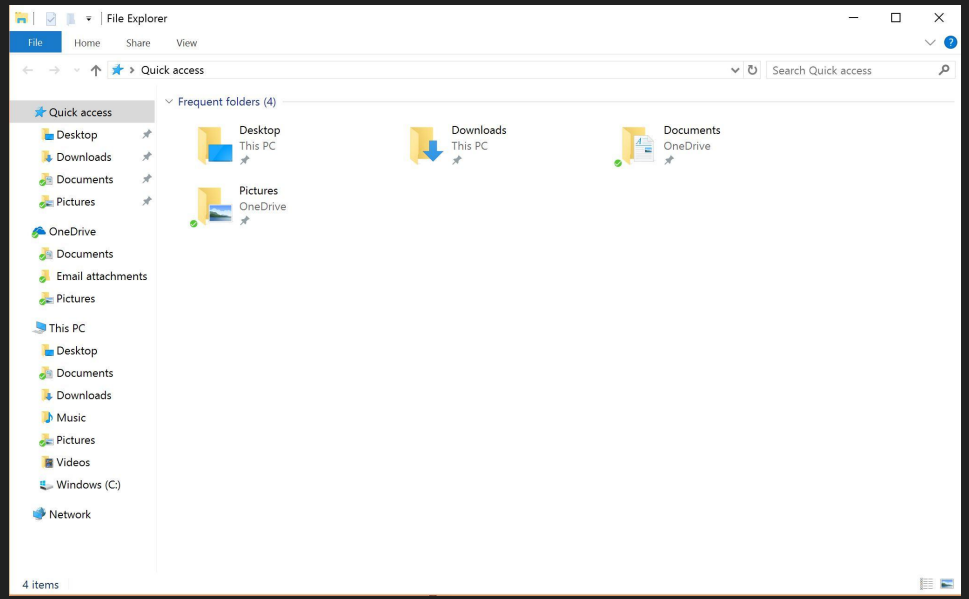


With windows 10, you can heavily customize the user interface better than the previous Windows OS. This can give you the ability to personalise it to your taste.

Windows 10 includes many personalization settings that let you change your desktop background, windows colours, lock screen background, and more.

To access this screen, you need to go to Personalization settings which is available at Settings > Personalization.

File Management

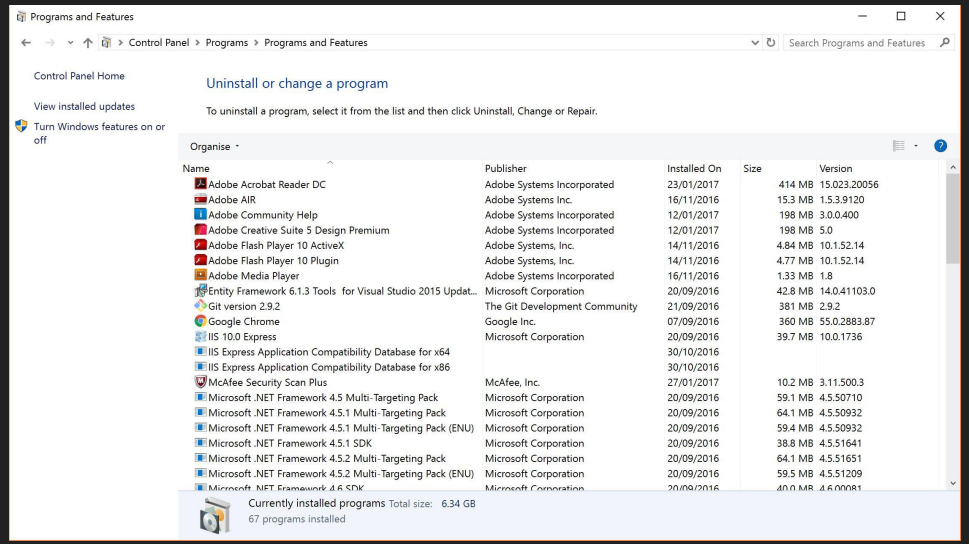


A critical part of Windows in general, is that you can organize files stored in the hard disk by using a File System that manages how files are stored and accessed on the computer.

To access File Explorer, you click Start > FIle explorer. When the File Explorer is launched, you get into the Quick access window, which pins the most frequently accessed folders.

You can use FIle Explorer for a variety of tasks. In addition to management and organization of files and folders, it’s also used to view and manage the resources of your computer such as internal storage, attached storage and optical drives.

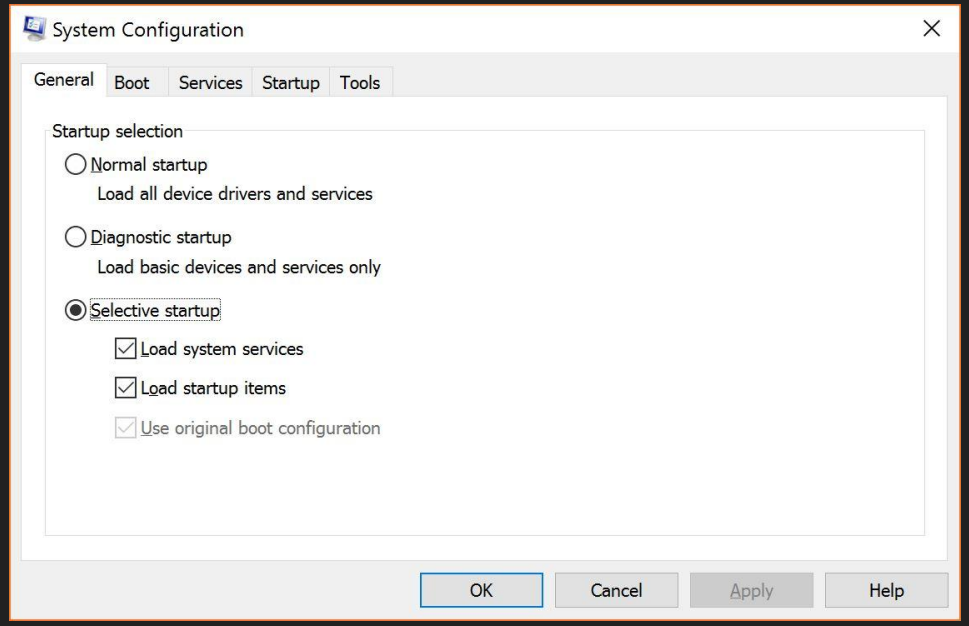
Application Management



Windows 10 offers multiple ways to access the applications. There are three ways in which you can manage and remove applications: Start Menu, Settings Window and from the Control Panel.

To open it from the start menu, you open the start menu and click on All Apps, and you will see a list of all the applications in alphabetical order. To open it from the Settings window, you search “settings” on the search bar and click on settings, click on system and choose Apps & features. To open it from the control panel, you type in “control panel” in the search bar, then you choose Programs and Features.

System Management Utilities

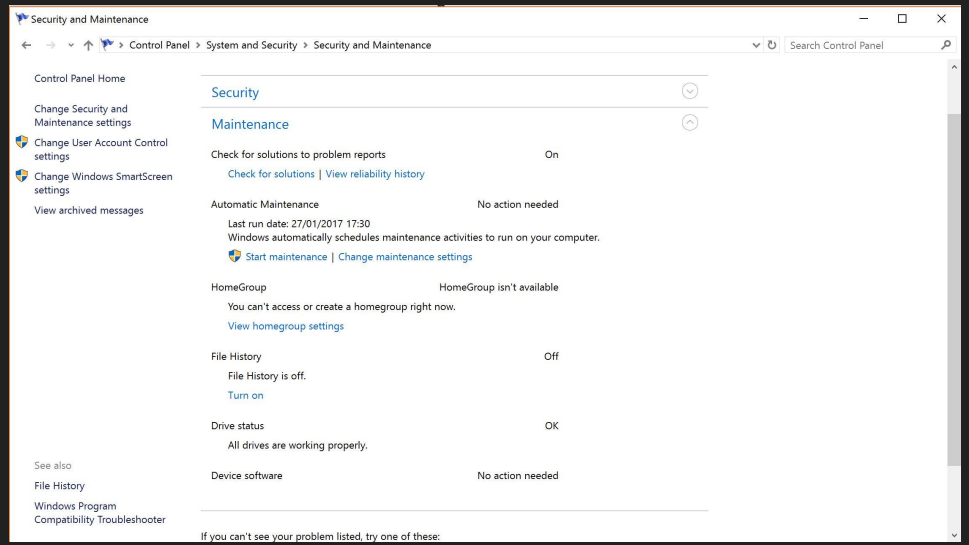


MSCONFIG is a utility built into WIndows to help with management and diagnostics of the Windows PC. By going on start > run window and typing “msconfig” a window will come up with the title “System Configuration”.

Once you're in the window, you can view all the background services that start when you boot your computer. You can also view all the programs that automatically start when you boot your computer.

Once you get inside these tabs, being able to turn items on and off will allow you to manage your memory usage and boot up times.

Windows Maintenance Tasks



Windows always had in-built tools that help customers to take care of their PC. Windows 10 has improved with the tools to make it a lot easier to take care of the PC. The automatic maintenance in WIndows 10 is the maintenance task on a windows 10 PC.

To open it you search for its settings via search box/cortana or through the control panel. Once open, you can set the time when the automatic maintenance runs its full diagnostic and cleanup. Choose a time when your computer is on and you're not using it for best results. You can also choose to let automatic maintenance in WIndows 10 turn off your PC to runs its process

Task 2 - Compare the features and functions of different operating systems

Leandro Baroca

Functions of Windows 7 & 10 Operating Systems

Kernel

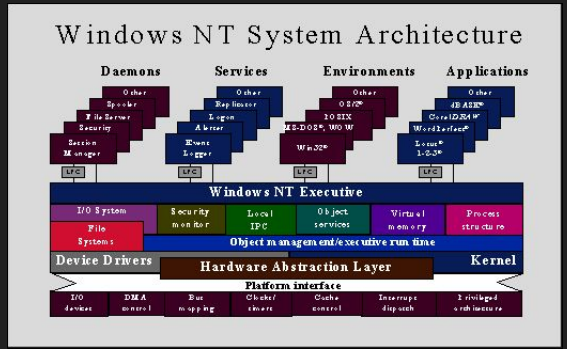
Windows 7

Windows 7 uses the Windows NT Kernel.

Windows 10

Windows 10 also uses the Windows NT Kernel.

Technically, there is no difference of the kernel between windows 10 and windows 7. The kernel in both windows are both the same kernels because they both contain the Windows NT Kernel, which has been used in Windows since Windows NT. The versions that the Windows NT kernel has been used on is: Windows NT 3.1, NT 3.5, NT 3.51, NT 4.0, 2000, XP, VIsta, 7, 8, 8.1 and 10.



User Interface

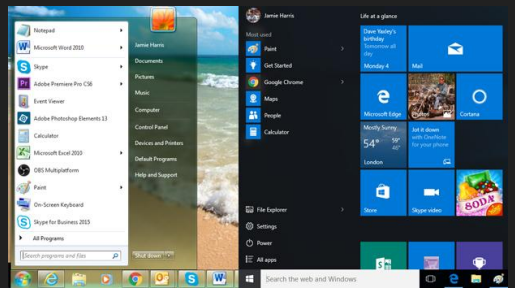
Windows 7

On Windows 7 you'll be used to clicking the Start Button and searching for content on your PC that way.

Windows 10

On Windows 10, it's all about Cortana, Microsoft's digital assistant which began life on Windows Phone.

The overall design of Windows 10 is the most striking difference to Windows 7. Everything has become flat, colourful and squared, unlike Windows 7 which was a bit glassy, rounded and used some gradients. Icons have also been simplified, but not beyond the point of recognition, again swapping the rounded look for something flat. With Windows 10, the search bar moved into the task bar with cortana added, instead of the start menu on Windows 7.



Device Drivers

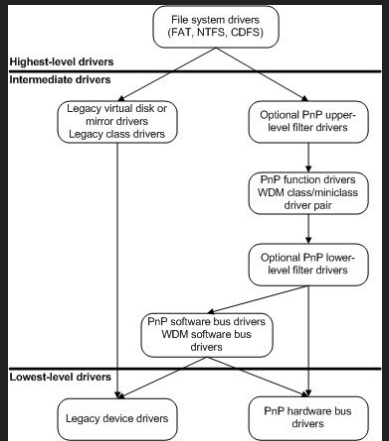
Windows 7

Windows 7 has a version of Microsoft WIndows Drivers that, if it hasn't been updated, would be outdated today

Windows 10

Since Windows 10 is a fairly new OS, it’s drivers are much more up to date than WIndows 7, but may still require updates.

Both drivers on the WIndows OS are the same type of drivers, so technically, there is no difference between the two. The only actual difference is the version of the driver to support the hardware that was popular of the time of the update.



File System Management

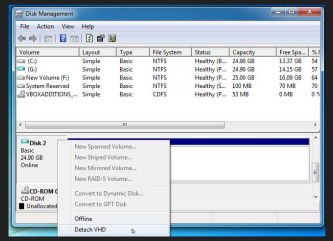
Windows 7

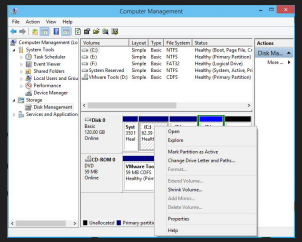
Windows 7 has a Disk manager with a glassy theme

Windows 10

Windows 10 has a Computer Management with a flat theme

There isn't much difference between WIndows 7 and Windows 10, but there are a few differences. Windows 7 has a different name of “Disk Manager” compared to Windows 10 “Computer Management”. Another difference is that Windows 10 has 4 different sections on the window compared to the 2 different sections in the windows 7. In Windows 10, the panel on the left is the navigation throughout the disk, which Windows 7 doesn’t have. The panel on the right on Windows 10 has actions which is used for more actions for the computer management, which Windows 7 doesn’t also have.





Registry

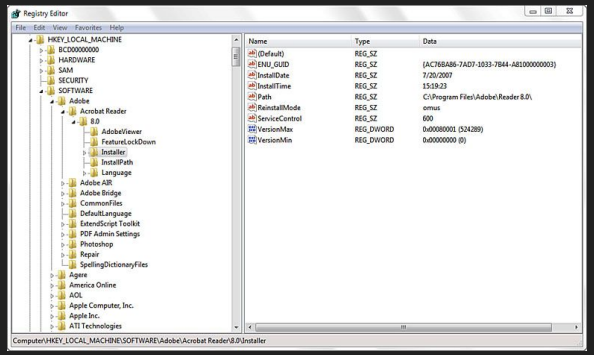
Windows 7

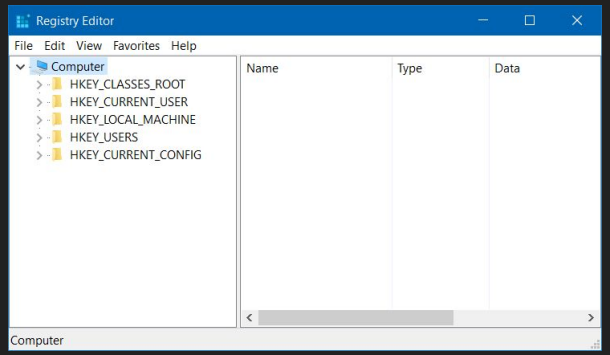
Has a Registry editor with the Windows 7 glassy theme

Windows 10

Has a Registry editor with the Windows 10 flat theme

The registry on both windows are also very similar visually because it is a very simplistic design that doesn't need any more additions, but there is a different aesthetic to both of them. The Windows 10 Registry has the “flat” theme which Windows 10 theme has, compared to the WIndows 7 “glassy” theme, but that is the only difference between the two.





Features of Windows 7 & 10 Operating Systems

Customizing the User Interface (Desktop, etc.)

Windows 7

With Windows 7, there is no way of heavily customising the UI like Windows 10, if there is any customisation at all. This is why Windows 7 looks similar on all machines, compared to the mass difference in other windows 10 machines.

Windows 10

As I mentioned in the last powerpoint, with windows 10, you can heavily customize the user interface better than the previous Windows OS. This can give you the ability to personalise it to your taste.

With workspace management, Windows 7 has multi-monitor support, though it’s somewhat limited with just access to either a mirroring mode or one that stretches the whole desktop across all the monitors, but with only one taskbar on one monitor. IT also has an early version of snap, the tool that automatically resizes windows when you drag them to the left or right.

With Windows 10 both these features have been enhanced. Different monitors can now have different backgrounds, with different slideshows too. Pictures can also now be set to span across all the monitors.

File Management

Windows 7

Windows 7 has a decent selection and certainly many people are used to them. However, they certainly aren’t the most advanced, and in some ways they’re a backwards step from what came before.

This one little feature summed up all that was wrong about the nonsensical path being trodden by Microsoft at the time. However, it wasn’t the only problem with Windows 7 File explorer. By any standard it was poorly designed, with wasted space and key functions hidden from view.

Windows 10

Windows 10 File Explorer and copy dialogue is much more useful, especially with addition in storage spaces. THis is a way to group multiple hard drives together to form a single logical drive.

While hardly revolutionary sounding, what makes it particularly useful is that it supports multiple levels of drive redundancy, to ensure that the data is backed up, and unlike a RAID system it’s completely hot-swappable and drives can be different sizes

Application Management

Windows 7

With Windows 7, there is only 1 way to manage applications, which is through the settings window.

This may seem like a good enough way of managing apps, but when compared to Windows 10, it cannot be accessed as easily or as versatile as Windows 10 is.

Windows 10

With Windows 10, there are many ways of managing applications. Which is via start menu, control panel and the settings window.

This makes it a lot easier for the user to see the apps and what the user can do with the apps depending on which method he is using to access the applications.

Basically, Windows 10 is much more faster for application management. For example, if you wanted to delete an application, the time it would take to do in Windows 10 would be significantly faster than Windows 7. This is due to the fact that you can do it just from the start menu in Windows 10 and for Windows 7 you needed to go to through the control panel to find the application management tool.

System Management Utilities

Windows 7

Windows 7 opens a window called “System Configuration”

Windows 10

Windows 10 also opens a window called “System Configuration”

With both Operating Systems, the MSCONFIG is very similar with the only difference being the theme. With both OS, it does the exact same thing of disabling and re-enabling software, device drivers or Windows services that run during startup process to help the user determine the cause of a problem with Windows.

The reason they are the same is that it only varies in Windows older than Vista. WIth VIsta and above, the tool gained additional support for launching a variety of tools, such as system information, other configuration areas, such as Internet options, and the ability to enable/disable UAC. Basically, there are no differences between the two.

Windows Maintenance Tasks

Windows 7

With Windows 7, It is opened via the control panel.

Windows 10

With Windows 10, It is also opened via the control panel.

As I mentioned before, Windows 10 has improved with the tools to make it a lot easier to take care of the PC. The automatic maintenance in WIndows 10 is the maintenance task on a windows 10 PC. What I meant by this is that Windows 10 has improved from the previous Windows OS because the PC can perform an automatic maintenance, which before you had to manually go through a wizard, like in windows 7, to choose what kind of maintenance you want your PC to carry out.

This means that a PC user that is less experienced would find Windows 10 much more advantageous for this because the PC does the work for you instead of you having to find out what kind of maintenance your PC needs.

Strengths and weaknesses

There’s nothing wrong with sticking with Windows 7, but upgrading to Windows 10 definitely has plenty of benefits, and not too many downsides.

Windows 10 has a lot of genuinely useful new features, as well as key improvements to old features, compared to Windows 7. None is enough on its own to make an upgrade necessary, but when combined they do make a compelling case. You’ve got better search, window management, file management, tools and more. Windows 10 is faster in general use too, and the new Start Menu is in some ways better than the one in Windows 7.

A downside to Windows 10 that Windows 7 doesn't have however, is not having control over Windows updates. Windows 10 completely removes the ability to manually control your system updates. Windows 10 Pro will let you defer updates for a few months, but that’s the extent of it. This isn’t much of a big deal for people who are stuck with Windows 7’s default Windows Update options, which downloaded patches automatically. But if you like to control your own update destiny, this could be a showstopper.

Unit 2 - Computer Systems
Assignment 4 - Operating Systems

Introduction

You work in a large computer store as a Technical Sales person and you advise customers and other members of staff on computer systems. A major part of your job is to give expert advice to customers as to which system to purchase based upon their specific needs. Another part of your job is to train staff members so that they can also advise customers as to the system specification to choose based upon their particular needs

Task 3 (2P3): Utility Software

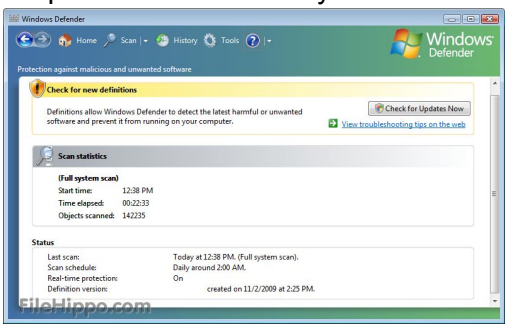
There are a number of software utilities that help the technician to maintain their systems complete the following table to explain the software utilities with a detailed example of its purpose.

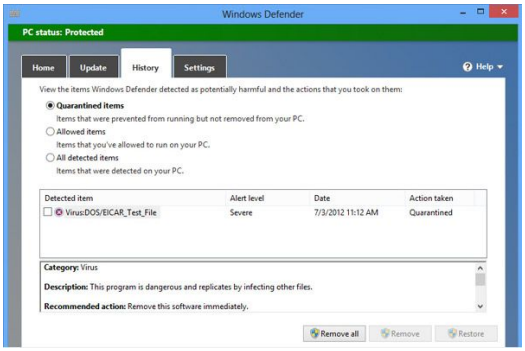
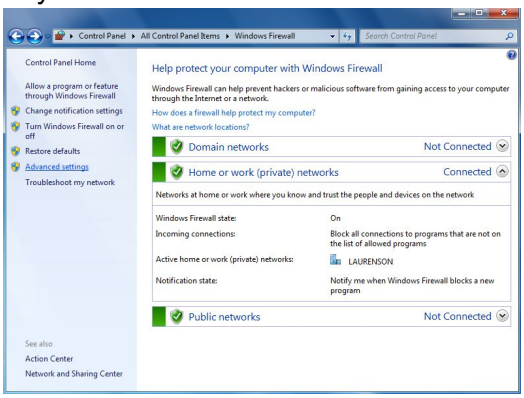
	Explain the Purpose of Utility	**Name of Windows Utility**	**Name of 3rd Party Utility**
Anti-virus Software	The Anti-Virus software is software that you can install on your computer so that you can be alerted when you have a virus. Anti-Virus software will help to keep the PC safe against worms, viruses, Trojan horses and other unwanted programs.	Windows Defender	Norton™ Antivirus
Firewalls	Firewalls, if properly used, can keep your computer safe from harm by setting up filters to detect and block intrusion attempts from external users such as hackers from the internet or any other network.	Windows Firewall	Norton Internet Security™
Clean up tools	Clean up tools helps free up space on the hard disk on the PC by searching the disk for files that you can safely delete. For example, you can choose to delete some or all of the files.	Disk Cleanup	CCleaner
Disk Defragmentation	Disk defragmentation is the process of putting file pieces back together. What disk defragmenters do is rewrite files into contiguous blocks of free space making sure that all file fragments are written in a consecutive order.	Disk Defragmenter	Auslogics Disk Defrag Professional

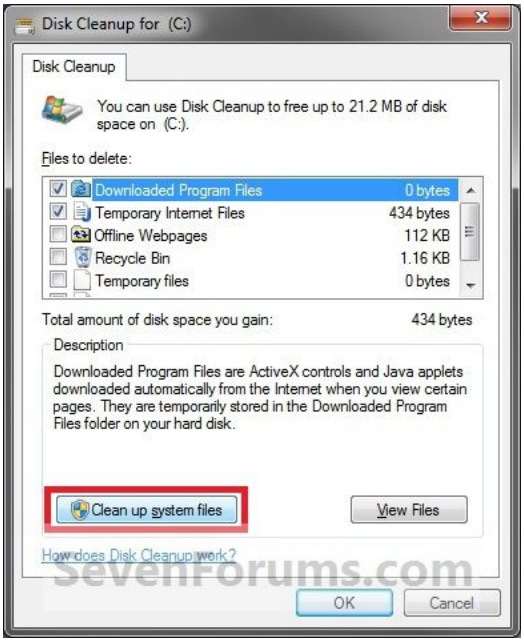
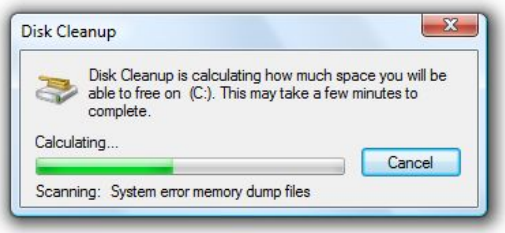
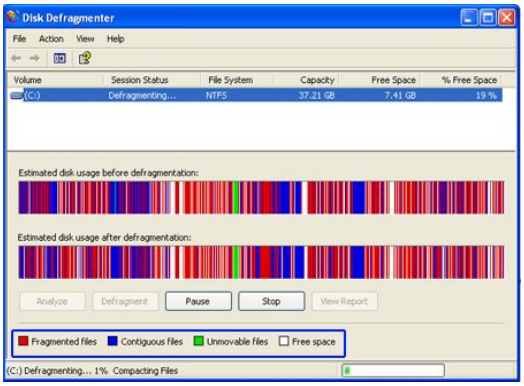
Disk Management	Disk management is used to manage the drives installed in a computer. For example, hard disk drives, optical disk drives and flash drives. Disk management can also be used to partition drives, format drives and assign drive letters.	Disk Management	Paragon Partition Manager™

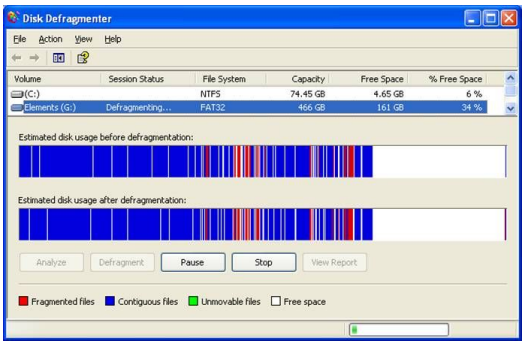
Task 4 (2D1):Use of Utility Software

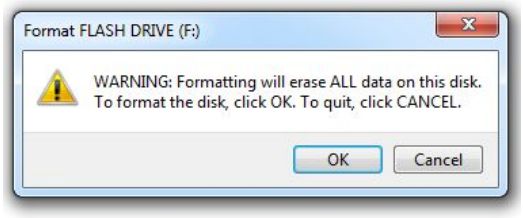
With reference to the utilities (Defragmentation, Disk Cleanup, etc.) identified in task 3 give 2 specific examples of how each utility can be used to improve to the performance of a system. Your detailed example should include reference to improvements in specific performance criteria (e.g. Access time, etc.)

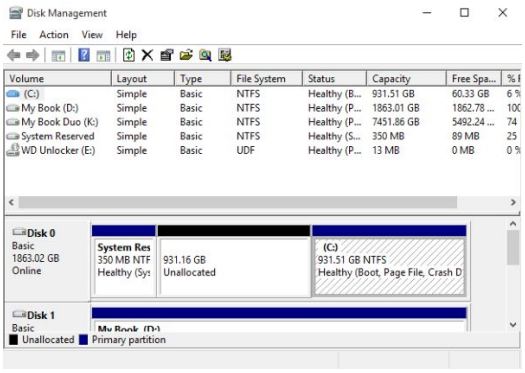
	Name of Windows/3rd Party Utility	**Explanation of how these utilities can improve the performance of a system (Illustrated with annotated screen shots)**
Anti-virus Software	Windows Defender The Image below is a screenshot of Windows Defender with the scan window open to search for malware or other harmful software which impact the performance of the system.  The image below is another anti virus software which has a table showing when the last scans were, this is important for system performance because this action has to be carried out regularly in order to keep system performance at its peak.	The reason that Virus protection improves the performance of a system, is that it maintain a safe environment for the user to use and it keeps out malware that will have the potential to slow down and decrease system performance and effectiveness. It does this by preventing malware from entering a system, detecting any malware that is on a system and ultimately removing that malware from the system. The second reason that the anti virus improves system performance is that it can free the system out of previously attained malware. If the system is malware free then the system can run effectively and without interference or performance decrease, this is because without virus protection malware would go relatively unnoticed on a system unless the user found and deleted the malware, but virus protection does this for the user and also speeds up

		the process as it is designed to perform this action.
Firewalls	Windows Firewall Below is a screenshot of the windows firewall showing that it is turned on and fully functional.  Below is an image of a pop up box of the firewall protecting the system from an unknown external program. 	The first reason that a firewall improves system performance is that this utility helps keep out unwanted traffic that can have the potential to harm the system with malware or malicious code, this helps to maintain a stable system performance by terminating any suspicious data packets that could contain malware or code that could damage the system. The second reason that firewalls improve system performance is that firewalls also stop the external programs, websites or users to steal valuable system information and files which would impact the performance or even corrupt the system itself, causing the system to shut down if such file like the “system32” file was removed.
Clean up tools	Disk Cleanup The image below is a screenshot of the Disk Cleanup menu with the button to start the cleaning process.	The first reason clean up tools improves system performance is that this utility deletes unnecessary files and data such as temp files, internet history, cookies and files in the systems recycling bin, this helps to free up space and keep the system’s

	 Below is a screenshot of a popup box of the Disk Cleanup clearing out the unwanted files of the computer system. 	performance from being decreased due to the excess of unused files. Deleting these unused and unnecessary files will help maintain a steady performance due to these files not slowing down the system. The second reason that this utility improves system performance is that it can also remove cookie files. Even though cookie files aren't a very significant problem to a computer system, but it can multiply at an alarming rate, especially if a user is using the internet frequently. Cookie files can prove to be dangerous when they aren't deleted constantly as that they can seriously impact a computer system's performance. To prevent this from happening, the cleanup tool that Windows has pre-installed will remove internet cookie files that are logged in a computer system, this should be done frequently due to the build up of cookies over time.
Disk Defragmentation	Disk Defragmenter Below is an image of the Disk Defragmenter showing the fragments of data within the disk drive. 	The first reason that Defragmentation improves system performance is that the utility frees up disk space and rearranges the fragmentations of data within a disk, which means that when trying to find a file the system can locate it easier than before. This increases system performance as it takes less time to find files and data. The second reason is due to the fact since defragmentation creates much more disk space which would improve the system by allowing it to run faster and also effectively extends the amount of data that can be stored on the storage media by rearranging the fragmentations and freeing up space.

<table>
<tr><td></td><td>Below is a screenshot of the Disk Defragmenter showing the disk drive after it has been properly defragmented and neatly set out, so the system can access the files much faster.
 
</td><td></td></tr>
<tr><td>Disk Management</td><td>Disk Management

Below is an image of a popup box showing if the user wants to format the F: drive of the system.
 

Below is a screenshot of the Disk Management showing all the disk drives within the pc, and they are all in healthy status.
 
</td><td>The first reason that disk management is good for improving the performance is that you can use it for drive formatting. What this is is basically to completely wipe out the data from a system, known as a factory reset. There are other methods of drive formatting which doesn't only consist of wiping the whole disk drive, you can have the option to only wipe one part of a hard drive, but it's advised to remove important information from the drive that is about to be formatted.

Another reason is that with this software you can remove certain programs and files that can have a massive impact on the performance of the computer system, such as software with large capacity or has a huge impact on boot up speed, which has a significant amount of impact to the performance of computer system.</td></tr>
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